

# Piyush Anand

Bangalore, India | thakur005piyush@gmail.com | +91-8340129180 | LinkedIn | GitHub | LeetCode

## SUMMARY

Data Science undergraduate with hands-on experience building end-to-end ML solutions — from data preprocessing and modeling to actionable business insights — across healthcare, agriculture, and automated ML tooling. Skilled in statistical analysis, machine learning, and data visualization, with a growing interest in full-stack development to turn models into deployable, user-facing products. Combines structured problem-solving with a strong drive to continuously learn new tools and technologies, aiming to bridge data science and software engineering in real-world applications.

## EDUCATION

**Bachelor's Degree, Computer Science & Engineering (Data Science)** — Bangalore Institute of Technology 2024 – 2028

- CGPA: **9.46/10.0** (as of Semester 4)
- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Database Management Systems
- Consistently maintained strong academic performance with a focus on data-driven problem-solving and analytical thinking
- Active in data science and machine learning projects alongside academic curriculum

## SKILLS

**Programming Languages (General Purpose):** Python, SQL, C, C++

**Core Concept:** Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, RAG, LangChain, HTML, CSS, OOP (Basics)

**Tools & Platforms:** Pandas, NumPy, Matplotlib, Scikit-learn, Git, GitHub, Jupyter Notebook, Docker

**Databases:** SQL, MySQL, MongoDB, PostgreSQL, Supabase

**GenAI & LLMs:** Prompt Engineering, LLM Application Development, OpenAI APIs, Agentic AI, Multi-Agent Systems

**Data Handling:** Data Cleaning, EDA, Data Preprocessing

## PROJECTS

**AUDAS – Autonomous AI Data Scientist** GitHub

- Architected a multi-agent AI system orchestrating 14+ specialist agents (profiling, cleaning, feature engineering, ML benchmarking and reporting) to autonomously convert raw datasets into production-ready ML projects from a natural-language goal.
- Built a FastAPI backend exposing REST APIs and real-time SSE log streaming, paired with a React + Vite frontend — deployed independently on Render and Vercel.
- Automated model selection by benchmarking 10+ algorithms (XGBoost, CatBoost, LightGBM, Random Forest, etc.) with leaderboard ranking and holdout metric evaluation.
- Enforced data integrity via strict train/val/test partitioning and leakage-guard validation; covered core pipeline logic with unit and integration test suites (Pytest).
- Integrated multi-provider LLM orchestration (Claude, GPT, Gemini, Groq, local Ollama) via LangChain for autonomous reasoning across the pipeline.
- *Tech Stack: Python, FastAPI, React, Vite, LangChain, Scikit-learn, XGBoost, Pytest, Render, Vercel*

**AgriSense AI – Agricultural Intelligence Platform** GitHub

- Built a production-grade full-stack platform (FastAPI + React/TypeScript) serving ML-driven crop recommendations across 22 crops via a versioned REST API (/api/v1) with Swagger documentation.
- Trained and benchmarked 5 ML algorithms for crop classification, selecting a tuned SVM model achieving 98.68% accuracy on standard validation and 89.20% on a 50,000-sample stress test.
- Designed a PostgreSQL (Supabase) schema with Row-Level Security across 13 tables, JWT authentication, and Redis caching for scalable multi-user access.
- Deployed backend on Render and frontend on Vercel, integrating live external APIs (weather and mandi market prices) with bilingual (English/Hindi) support.
- *Tech Stack: Python, FastAPI, PostgreSQL, Supabase, Redis, Docker, React, TypeScript, Pytest, Render, Vercel*

**SwasthSetu – Care Continuity Platform** GitHub Hackathon 2025

- Collaborated in a 3-member team to build a role-based healthcare workflow platform for reception and doctor staff, powered by 7 REST API endpoints across authentication, patient search, consent, and records.
- Implemented OTP-based consent verification and a route-protected role system (reception/doctor) with session-cookie authentication.
- Designed an emergency-first, FIFO-fallback triage queue with real-time doctor assignment across 6 medical specialties.
- Built a data-access layer with MongoDB persistence and automatic in-memory fallback, ensuring the demo runs reliably without external infrastructure.
- *Tech Stack: Next.js, React, TypeScript, MongoDB*

## CERTIFICATIONS

- |                                                                    |                |          |
|--------------------------------------------------------------------|----------------|----------|
| • <b>Google AI Essentials</b> — Google via Coursera                | May 2026       | Verify ↗ |
| • <b>Data Analytics Job Simulation</b> — Deloitte                  | September 2026 | Verify ↗ |
| • <b>Ideatattva IDEATHON</b> — Nikshatra E-summit                  | Nov 2025       | Verify ↗ |
| • <b>Understanding Agentic AI</b> — Digital Workforce Services Plc | May 2026       | Verify ↗ |